King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam I Term 242

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The projection of $\vec{\mathbf{u}} = \langle 0, 2, -3 \rangle$ onto $\vec{\mathbf{v}} = \langle 1, -1, 2 \rangle$ is

(a) $\left\langle \frac{1}{3}, -\frac{1}{3}, \frac{1}{3} \right\rangle$

(b) $\left\langle -1, 1, -\frac{2}{3} \right\rangle$

(c) $\left\langle -\frac{4}{3}, \frac{4}{3}, -\frac{8}{3} \right\rangle$

(d) $\left\langle 0, -\frac{8}{3}, 4 \right\rangle$

(e) $\left\langle \frac{2}{3}, -\frac{2}{3}, \frac{4}{3} \right\rangle$

2. Which one of the following statements is TRUE: ($\vec{\mathbf{u}}$, $\vec{\mathbf{v}}$, and $\vec{\mathbf{w}}$ are vectors in 3D-space, $\vec{\mathbf{o}}$ is the zero vector)

(a) $(\vec{\mathbf{u}} + \vec{\mathbf{v}}) \times (\vec{\mathbf{u}} - \vec{\mathbf{v}}) = 2(\vec{\mathbf{v}} \times \vec{\mathbf{u}})$

(b) $\|\vec{\mathbf{u}} + \vec{\mathbf{v}}\|^2 = \|\vec{\mathbf{u}}\|^2 + \|\vec{\mathbf{v}}\|^2$

(c) If $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} = \vec{\mathbf{u}} \cdot \vec{\mathbf{w}}$ and $\vec{\mathbf{u}} \neq \vec{\mathbf{o}}$, then $\vec{\mathbf{v}} = \vec{\mathbf{w}}$

(d) $\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} + \vec{\mathbf{u}} \times \vec{\mathbf{v}} = \vec{\mathbf{v}} \cdot \vec{\mathbf{u}} - (-\vec{\mathbf{v}}) \times \vec{\mathbf{u}}$

(e) If $\vec{\mathbf{u}} \neq \vec{\mathbf{o}}$ and $\vec{\mathbf{v}} \neq \vec{\mathbf{o}}$, then $\vec{\mathbf{u}} \times \vec{\mathbf{v}} \neq \vec{\mathbf{o}}$

3. The parametric curve

$$x = 2 + t^2, \quad y = 1 + t + t^3$$

is

(a) concave downward on $\left(-\frac{1}{\sqrt{3}}, 0 \right)$

(b) concave downward on $\left(0, \frac{1}{\sqrt{3}} \right)$

(c) concave downward on $(0, \infty)$

(d) concave upward on $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$

(e) concave upward on $(0, \infty)$

4. If the points

$$P(1,-2,k), \quad Q(k-1,1,0), \quad R(4,7,-2k)$$

are collinear, then $k^3 - 2k + 10 =$

(a) 0

(b) -15

(c) -21

(d) 29

(e) 31

5. The area of the region lying in the first quadrant and inside the polar curve $r^2 = 2\cos(2\theta)$ is equal to

(a) $\frac{1}{4}$

(b) $\frac{1}{2}$

(c) 1

(d) $\frac{1}{\sqrt{2}}$

(e) $\sqrt{2}$

6. The length of the parametric curve

$$x = \frac{1}{3}t^3 - t, \quad y = t^2, \quad 0 \le t \le 3$$

is equal to

(a) 15

(b) $\frac{14}{3}$

(c) $\frac{5}{3}$

(d) 12

(e) 9

7. In the xy-plane, if the vector $\vec{\mathbf{u}}$ is of magnitude 2 and makes an angle of $\frac{\pi}{3}$ with the positive x-axis, and the vector $\vec{\mathbf{u}}+\vec{\mathbf{v}}$ is of magnitude $2\sqrt{3}$ and makes an angle of $\frac{\pi}{2}$ with the positive x axis, then

(a) $\vec{\mathbf{v}} = \langle 1, \sqrt{3} \rangle$

(b) $\vec{\mathbf{v}} = \langle 0, -\sqrt{3} \rangle$

(c) $\vec{\mathbf{v}} = \langle -1, \sqrt{3} \rangle$

(d) $\vec{\mathbf{v}} = \langle 1, 2\sqrt{3} \rangle$

(e) $\vec{\mathbf{v}} = \langle -1, 0 \rangle$

8. Let $x > 0$. If the volume of the parallelepiped with adjacent edges

$$\vec{\mathbf{u}} = \langle 2, x, -x \rangle, \quad \vec{\mathbf{v}} = \langle 3, x, 4 \rangle, \quad \vec{\mathbf{w}} = \langle 5, -3, 3 \rangle$$

is equal to 55, then $x^3 + x^2 - 4 =$

(a) 2

(b) 3

(c) 0

(d) -5

(e) -2

9. If $\vec{\mathbf{u}} = \langle a, b \rangle$, with $a < 0$, is a unit vector that is parallel to the tangent line to the curve $y = -x^3 + x^2 - 1$ at the point $(2, -5)$, then $ab =$

(a) $-\frac{4}{63}$

(b) $\frac{2}{65}$

(c) $\frac{8}{65}$

(d) $\frac{8}{63}$

(e) $-\frac{8}{65}$

10. The area of the triangle with vertices

$$P(2,2,2), \quad Q(5,4,0), \quad R(-1,1,1)$$

is equal to

(a) $\dfrac{\sqrt{97}}{2}$

(b) $\dfrac{\sqrt{106}}{2}$

(c) $\dfrac{3\sqrt{10}}{2}$

(d) 5

(e) $\dfrac{\sqrt{101}}{2}$

11. The slope of the tangent line to the polar curve

$$r = 2 + \cos(3\theta)$$

at the point corresponding to $\theta = \dfrac{\pi}{2}$ is equal to

(a) $-\dfrac{3}{2}$

(b) 3

(c) -3

(d) $\dfrac{1}{2}$

(e) -1

12. The graph of the parametric curve

$$x = \sin t, \quad y = \cos(2t), \quad t \in (-\infty, \infty)$$

is

(a) a hyperbola

(b) an ellipse

(c) part of a line

(d) a full parabola

(e) part of a parabola

13. The polar equation $\theta = \dfrac{5\pi}{6}$ can be converted to the rectangular equation

(a) $y = -\dfrac{\sqrt{3}}{3}x$

(b) $y = \dfrac{\sqrt{3}}{2}x$

(c) $y = \sqrt{3}\,x$

(d) $y = -2\sqrt{3}\,x$

(e) $y = -\sqrt{3}\,x$

14. The parametric curve

$$x = t^2 - 2t + 1, \quad y = t^3 - 6t^2$$

has a vertical tangent line at the point

(a) $(x, y) = (9, -32)$

(b) $(x, y) = (1, -16)$

(c) $(x, y) = (4, -7)$

(d) $(x, y) = (1, 0)$

(e) $(x, y) = (0, -5)$

15. The area of the region in the common interior of the polar curves $r = 2(1+\cos\theta)$ and $r = 2\sin\theta$ is equal to

(a) $2\pi + 3$

(b) $2\pi - 4$

(c) $\pi + 6$

(d) $2\pi + 5$

(e) $2\pi - 2$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\frac{-8}{6}\langle 1,-1,2\rangle = \langle -\frac{4}{3},\frac{4}{3},-\frac{8}{3}\rangle$.
2	**(a)**	$(\mathbf{u}+\mathbf{v})\times(\mathbf{u}-\mathbf{v}) = 2(\mathbf{v}\times\mathbf{u})$; the others have counterexamples.
3	**(b)**	$y''_{xx} = \frac{3t^2-1}{4t^3} < 0$ for $0 < t < \frac{1}{\sqrt{3}}$.
4	**(e)**	$\overrightarrow{PR} = 3\overrightarrow{PQ} \Rightarrow k = 3$, so $27 - 6 + 10 = 31$.
5	**(b)**	$A = \int_0^{\pi/4} \cos 2\theta \, d\theta = \frac{1}{2}$.
6	**(d)**	$\sqrt{(t^2-1)^2+4t^2} = t^2+1$, $L = \int_0^3 (t^2+1)dt = 12$.
7	**(c)**	$\mathbf{u} = \langle 1,\sqrt{3}\rangle$, $\mathbf{u}+\mathbf{v} = \langle 0, 2\sqrt{3}\rangle$, so $\mathbf{v} = \langle -1,\sqrt{3}\rangle$.
8	**(e)**	$\lvert \det \rvert = 5x^2+26x+24 = 55 \Rightarrow x = 1$, so $1+1-4 = -2$.
9	**(e)**	Slope -8, $\mathbf{u} = \frac{1}{\sqrt{65}}\langle -1, 8\rangle$, so $ab = -\frac{8}{65}$.
10	**(b)**	$\overrightarrow{PQ}\times\overrightarrow{PR} = \langle -4,9,3\rangle$, area $= \frac{1}{2}\sqrt{106}$.
11	**(a)**	$r = 2$, $r' = 3$ at $\frac{\pi}{2}$: slope $= \frac{3}{-2} = -\frac{3}{2}$.
12	**(e)**	$y = 1-2x^2$ with $-1 \le x \le 1$: part of a parabola.
13	**(a)**	$\frac{y}{x} = \tan\frac{5\pi}{6} = -\frac{1}{\sqrt{3}}$, so $y = -\frac{\sqrt{3}}{3}x$.
14	**(e)**	$x' = 2t-2 = 0 \Rightarrow t = 1$, $y'(1) = -9 \neq 0$: point $(0,-5)$.
15	**(b)**	$\frac{\pi}{2}$ (circle part) $+ \left(\frac{3\pi}{2} - 4\right)$ (cardioid part) $= 2\pi - 4$.